INSPIRE INTELLIGENCE
Inspire Intelligence Journal
ISSN: 3110-942X
InspireQuill.org
Inspire Academic Press (Canada)

Research Paper

A Graph Neural Network and Reinforcement Learning Framework for Resilient Self-Healing Routing in Disaster-Aware Healthcare Wireless Sensor Networks

Ibrahim Salman[1], Jaspreet Kaur[1]

[1] Computing Science Department, Thompson Rivers University, BC, Canada

*** Corresponding Author: [TO BE CONFIRMED WITH PROF. JASPREET BEFORE JOURNAL SUBMISSION]**

Abstract

Healthcare Wireless Sensor Networks (HWSNs), heavily used in the industry for continuous remote patient monitoring, face severe reliability challenges during disaster scenarios, where energy depletion, environmental hazards, interference, and congestion drive frequent node failures. Existing routing approaches, including shortest-path, energy-aware, and traditional fault-tolerant schemes, are often reactive, adapting only after disruption has already occurred. This paper presents a self-healing routing framework for disaster-aware HWSNs that combines a Graph Neural Network (GNN) with Reinforcement Learning (RL). The GNN estimates each node's failure risk using both node-level features and the surrounding network topology, and a Q-learning routing agent uses that risk estimate to choose more reliable paths. The framework was evaluated in simulation against several baselines, including shortest-path, energy-aware, weighted failure-risk-aware, LSTM-based failure prediction, and LSTM-RL self-healing methods. Tests were run under four disaster severity levels (Low, Medium, High, and Extreme) with 30 to 100 trials per condition. An ablation study was used to isolate the GNN's role in the framework. When trained as a binary classifier, the GNN failed to outperform the simple weighted-formula baseline. Retraining the GNN through regression closed the gaps in latency, hop count, and energy cost, although formula-based routing retained an advantage in disaster-risk exposure. Under Extreme disaster severity, the proposed framework also achieved a statistically significant improvement in route delivery ratio over LSTM-RL, reaching 96.0% compared with 89.6% (paired t-test, $p = 0.027$, $n = 100$). In several Extreme-severity trials, GNN-RL found valid routes for most or all patient nodes where LSTM-RL failed entirely, while also using roughly 15 times fewer trainable parameters. Overall, the results suggest that topology-aware failure prediction appears to provide a real resilience advantage in extreme disaster scenarios, though this benefit was not consistent across every severity level. They also show that architecture alone is not enough; the way training labels are defined also matters. Future work should validate the framework on real-world datasets with labeled node failures.

Keywords: Healthcare Wireless Sensor Networks; Graph Neural Networks; Reinforcement Learning; Self-Healing Routing; Disaster-Aware Networks; Failure Prediction

1. Introduction

Continuous patient monitoring that integrates wearable sensors, body-area networks, and wireless communication technologies to transmit physiological data to healthcare providers in real time increasingly relies on Healthcare Wireless Sensor Networks (HWSNs) [1]. In this context, communication reliability is essential rather than secondary, because delayed or missing readings can directly affect patient assessment and emergency intervention [3].

In critical natural disasters, this becomes essential as natural hazards, infrastructure failures, and large-scale emergencies can disrupt network operation. This can happen through node failures, interference, and deteriorating communication links [6], often when continuous monitoring is needed most. Routing protocols designed for stable network conditions are generally not equipped to respond effectively to this level of disruption.

A large portion of the existing Wireless Sensor Network (WSN) routing literature focuses on minimizing hop count, reducing energy consumption, or extending network lifetime [11]. However, these objectives do not fully address the requirements of disaster environments. Shortest-path routing can forward packets efficiently, but it does not account for node reliability or environmental danger. Energy-aware routing improves battery usage by avoiding low-energy nodes, but it does not consider communication quality or the probability of imminent node failure [8]. Fault-tolerant routing methods address reliability more directly, although many remain reactive and initiate route recovery only after a path has already failed [6], [7]. This delay is itself a serious limitation, because even brief communication interruptions may result in lost or delayed medical information.

Machine learning provides a partial solution to this problem. Long Short-Term Memory (LSTM) networks have shown potential for identifying temporal patterns associated with node degradation [12]–[14]. However, LSTM-based methods are primarily designed to model changes over time rather than relationships within a network structure. As a result, they often treat node failures as largely independent events, even though WSNs are graph-structured systems in which a node's condition may be strongly influenced by neighboring nodes.

Graph Neural Networks (GNNs) are well suited to address this structural limitation because they learn node representations by combining local node features with connectivity information [16]–[20]. Despite their increasing use in wireless communication and Internet of Things applications [20]–[24], relatively few studies have applied GNNs to topology-aware failure prediction for disaster-aware healthcare routing. At the same time, Reinforcement Learning (RL) has demonstrated the ability to learn adaptive routing policies that balance multiple objectives through long-term reward rather than fixed decision rules [25]–[31]. What remains largely unexplored is an adaptive routing agent that responds to predicted failure risk instead of relying only on current network conditions.

To address this gap, this study develops a routing framework that combines a Graph neural Network with Q-learning. The GNN evaluates the likelihood of node failure risk using both node-state information and network topology, while a Q-learning agent uses these predictions to make proactive routing decisions. Importantly, the study also presents a rigorous ablation analysis. The initial GNN implementation did not outperform a simple hand-crafted risk formula, and this limitation is examined and corrected rather than omitted in favor of reporting only the final successful configuration.

The main contributions of this work are as follows:

- A GNN-based failure-risk model incorporating both node-level conditions and topological context evaluated statistically across 30 –100 trials over four disaster severity levels.
- A Q-learning routing agent that incorporates GNN-predicted failure risk into its reward calculation to support proactive route selection.
- An ablation study that isolates the GNN's specific contribution, showing that classification-based training does not improve upon a hand-crafted formula, followed by a corrected regression-based training approach that closes most of this performance gap.
- A demonstration that the corrected framework delivers more reliable routing under Extreme disaster severity, while using far less computational power than the LSTM-based prediction.

The remainder of this paper is organized as follows. Section 2 reviews related literature, Section 3 presents the proposed framework, Section 4 describes the methodology, Section 5 reports the experimental results (including the ablation study), Section 6 discusses the findings, Section 7 outlines the limitations and future work, and Section 8 concludes the paper.

2. Literature Review

Healthcare Wireless Sensor Networks (HWSNs) support continuous patient monitoring by collecting physiological data through wearable, implantable, and environmental sensors. HWSNs are used in developing applications such as remote patient care, chronic disease monitoring, and emergency response, where timely flow of medical data allows healthcare professionals to assess a patient's condition. This helps

professionals respond, when necessary [1], [2]. Medical data are often time-sensitive, and communication failures can negatively affect patient monitoring and clinical decision-making [3 Several studies have proposed energy-efficient and quality-of-service-aware routing protocols for healthcare sensor networks][4]. Although these approaches improve network lifetime and communication performance under stable conditions, they do not fully address large-scale disruptions caused by disasters, infrastructure damage, or environmental hazards.

Disaster-resilient WSNs are designed to maintain network functionality when nodes or communication links become unreliable due to environmental disturbances, physical damage, or congestion [5]. Such capabilities are particularly important in healthcare monitoring systems deployed during emergency response operations, natural disasters, and post-disaster medical assistance. Traditional fault-tolerant routing protocols typically depend on backup routes, route rediscovery, or link-quality monitoring to recover from failures [6]. Recent studies have argued for proactive routing strategies that anticipate failures before they disrupt communication [7]. However, existing disaster-aware routing methods often rely on manually designed risk metrics or predefined rules that may not generalize effectively across different network conditions.

Machine learning has increasingly been applied to WSNs for anomaly detection, energy prediction, and fault diagnosis [12]. Long Short-Term Memory networks are designed to learn temporal dependencies in sequential data [13], [14]. This makes them potentially suitable for modeling changes in node characteristics such as residual energy, packet loss, signal strength, and communication delay. Their primary strength is their ability to learn how node conditions evolve over time, providing early warning ahead of failure.

However, they treat network entities as independent observations and do not account for dependencies between neighboring nodes, so temporal modeling alone captures only part of how failures propagate in WSNs.

GNNs address this structural gap by learning jointly from node features and graph topology [15], [16], making them naturally suited to wireless networks and IoT infrastructures. Architectures such as Graph Convolutional Networks [18], Graph Attention Networks [19], and GraphSAGE [20] have advanced this capability through message-passing. GNNs have been applied to resource allocation, network optimization, and link-quality estimation in wireless settings [21]-[25], but their application to topology-aware node failure prediction in healthcare-focused, disaster-resilient routing remains limited. Most existing studies focus on network optimization or traffic analysis rather than proactive self-healing communication mechanisms.

Where GNNs address structural prediction, Reinforcement Learning addresses the routing decision itself. An agent learns routing policies through rewards and penalties tied to packet delivery, energy use, latency, and reliability [25], building on foundational algorithms such as Q-learning [27], deep Q-networks [28], proximal policy optimization [29], and actor-critic methods [30]. RL-based routing can outperform static routing approaches in dynamic environments because it adapts to changing network conditions and handles multiple routing objectives more flexibly than fixed decision rules [10], [31], [32]. However, most RL-based routing operates on current network observations alone, without predictive failure information, so these agents adapt only once performance degradation is already visible.

These research areas, healthcare WSN routing, disaster-resilient routing, LSTM-based prediction, GNN-based representation learning, and RL-based adaptive routing, have largely evolved independently. The framework proposed in this paper integrates topology-aware failure prediction with adaptive routing and unlike most prior studies, verifies the GNN's actual contribution through an ablation study rather than assuming its effectiveness. This methodological step proved essential, since the initial results showed no performance advantage.

3. Proposed GNN-RL Self-Healing Routing Framework

3.1 Framework Overview

The proposed framework combines Graph Neural Network (GNN)-based failure prediction with Reinforcement Learning (RL)-based adaptive routing to support reliable communication in disaster-aware HWSNs. Unlike conventional routing, which responds only after a node or link has failed, the framework evaluates the network state before routing and assigns node-risk estimates that guide subsequent routing decisions. These predictions feed into a Q-learning agent, which learns to select reliable paths while balancing latency, energy consumption, and disaster exposure.



Figure 1. Architecture of the proposed GNN-RL self-healing routing framework.

3.2 Network Model

The network is modeled as an undirected graph G = (V, E), where V is the set of nodes (patient sensors, relay nodes, and a sink) and E the set of communication links. A multi-hop model is assumed, since direct sensor-to-sink communication is infeasible due to range and energy constraints. A disaster zone is introduced into the topology with nearby nodes experiencing increased degradation and higher failure risk.

3.3 Node-State Representation

Each node is represented by a six-dimensional feature vector: $X_i = [E_i, PLR_i, RSSI_i, D_i, FL_i, R_i]$ — residual energy, packet loss rate, RSSI, communication delay, forwarding load, and disaster risk level, selected based on prior evidence linking them to node reliability [6], [8], [9].

3.4 GNN-Based Risk Prediction

Failures in wireless sensor networks often exhibit spatial structure, particularly during disasters, when neighboring nodes degrade together rather than independently. A GNN addresses this by learning node representations jointly from feature information (matrix X) and graph connectivity (matrix A) through message-passing [16]–[18]. At each propagation layer l, node i aggregates information from its neighborhood:

$$h_i^{(l)} = \sigma(W^{(l)} \cdot \text{AGGREGATE}(\{h_j^{(l-1)} : j \in N(i)\} \cup \{h_i^{(l-1)}\}))$$

The final embedding produces a continuous risk score:

$$P_failure(i) = sigmoid(w_{out}^{T} \cdot h_i^{(L)} + b_{out})$$

A critical methodological finding of this work concerns the training objective for this output. An initial implementation trained the GNN via binary cross-entropy loss against thresholded pseudo-labels:

$$L_GNN(BCE) = -(1/|V|) \Sigma_i [y_i \log(P_failure(i)) + (1-y_i) \log(1-P_failure(i))]$$

where $y_i \in \{0,1\}$ is a binary label derived from a weighted failure-risk formula (Section 4.6). Section 5.3 shows that this classification-based training reached high accuracy, but its risk predictions did not match the formula's actual values closely. This was a problem because the RL reward function depends on those exact values. The GNN was therefore retrained using mean-squared-error regression directly against the formula's continuous output r_i:

$$L_GNN(MSE) = (1/|V|)\,\Sigma_i\ (P_failure(i) - r_i)^2$$

This regression-based model is the one used in the final results (Sections 5.4–5.5).

3.5 Reinforcement Learning-Based Adaptive Routing

Routing is formulated as a sequential decision-making problem [26], [30] and solved using Q-learning. The current node state, while its connected neighbors define the available actions. Each next-hop decision is rewarded based on progress toward the sink, GNN-predicted risk, latency, energy cost, link quality, and hop efficiency, balancing successful delivery against these costs. Reaching the sink earns a large positive reward, while selecting a high-risk node brings a substantial penalty. Training is performed over repeated episodes using standard Q-learning updates, with epsilon-greedy exploration gradually reduced as training progresses.

3.6 Self-Healing Routing Mechanism

The framework's self-healing capability arises from the interaction between the GNN and the RL agent. Node-state information is collected, and the GNN estimates a continuous failure-risk score for every node which is then assigned to the corresponding nodes in the network graph. The RL agent uses these scores when evaluating possible next-hop actions. Routes through higher risk nodes receive lower rewards, encouraging the agent to select safer alternatives. As a result, the routing process can adapt proactively without waiting for a node to fail before initiating route recovery.

3.7 Framework Summary

The final framework uses regression-based GNN risk estimates during Q-learning route selection. Ablation experiments showed that this design choice matters: the original classification model underperformed compared with direct formula-based baselines, while regression closed most of that performance gap. The clearest benefit over LSTM-RL appeared under Extreme disaster severity.

4. Methodology

4.1 Research Methodology Overview

This study investigates whether combining topology-aware failure prediction with adaptive routing improves communication reliability in disaster-aware HWSNs. It also examines whether the GNN component's contribution can be demonstrated through an ablation study rather than simply assumed. A simulation-based methodology was used, in which the proposed framework was implemented and compared with five baseline approaches under identical network conditions. The evaluation covered four disaster severity levels and included repeated trials and paired statistical testing.

4.2 Simulation Environment

The implementation was built in Python, using NetworkX for topology generation, PyTorch for GNN and LSTM implementation, and NumPy/Pandas/SciPy for data handling and statistical testing.

4.3 Network Configuration

The simulated network consists of 74 nodes: 8 patient sensor nodes, 65 relay nodes, and 1 sink. These nodes are distributed across a (100 x 100)-unit area, with a disaster zone of radius 28 units located at the center of the network. The patient nodes are positioned far from the sink, requiring packets to be forwarded through multiple relay nodes using multi-hop routing.

4.4 Disaster Severity Levels

Four severity levels: Low, Medium, High, and Extreme were modeled by scaling environmental degradation parameters (temperature/disaster risk, packet loss, delay, energy drain, forwarding load, signal

strength) applied to nodes within and near the disaster zone, with Extreme representing approximately a 1.5× scaling beyond High across all parameters.

4.5 Baseline Routing Models

Five baseline approaches were implemented. Shortest-Path selects routes with the fewest hops, while the Energy-Aware avoids nodes with low residual energy. Weighted Failure-Risk-Aware baseline uses a manually defined formula combining six node features with fixed weights: 0.25 for energy risk, 0.25 for disaster risk, 0.20 for packet loss, and 0.10 each for RSSI, delay, and forwarding-load risk. The remaining two baselines are LSTM-Based Failure Prediction and LSTM-RL Self-Healing Routing, which integrates LSTM-based prediction with Q-learning and serves as the primary intelligent baseline.

4.6 GNN Training

Both GNN variants—the classification-based model and the regression-based model—were trained for 250 epochs using the Adam optimizer with a learning rate of 0.01. Because real-world failure-labeled data was unavailable, the training labels and targets were derived from the weighted-risk formula described in Section 4.5. For the classification variant, the formula outputs were converted into binary labels using a threshold of 0.55. In contrast, the regression variant used the formula's continuous risk values directly as the training targets.

4.7 Q-Learning Configuration

The routing agent was trained for 1,500 episodes with learning rate of 0.12, a discount factor of 0.85, and an initial exploration rate 0.35 decaying by 0.997 per episode to a minimum of 0.03 a maximum of 18 routing steps per episode.

4.8 Implemented Regression-Based GNN-RL Routing Procedure

Algorithm: Regression-Based GNN-RL Routing Procedure
Input: WSN graph G, source node s, sink node d
Output: Selected route from s to d
1. Generate six node features.
2. Construct and normalize the adjacency matrix.
3. Generate continuous weighted-risk targets.
4. Train the regression GNN using MSE loss.
5. Predict and assign failure-risk scores to all nodes.
6. Initialize the Q-table.
7. Train the Q-learning agent for 1,500 episodes, selecting actions via epsilon-greedy exploration and updating Q-values using rewards based on progress, risk, latency, energy, and link quality.
8. After training, select the highest-valued valid next hop.
9. Continue until the sink is reached or no valid route remains.
10. Return the final route.

In the implemented prototype, GNN risk scores are calculated before Q-learning begins and remain fixed during each experiment. They are not recomputed while an individual route is active. For the ablation study, the same Q-learning procedure is evaluated using either the weighted-formula risk directly or the GNN-predicted risk.

4.9 Ablation Study Design

To isolate the GNN's specific contribution, two routing conditions were evaluated with RL agent, while keeping the network configuration, and all other parameters unchanged. The first condition used the weighted-formula risk values directly within the RL routing process, without involving a GNN. The second used GNN-predicted risk values and represented the proposed model. This comparison was repeated over 30 trials at Low, Medium, and High disaster severity, with a different randomly generated network realization used in each trial.

4.10 Extreme Severity Evaluation

Given the ablation's diagnostic value, the corrected regression-based GNN-RL model was additionally evaluated against LSTM-RL specifically under Extreme severity across 100 independent trials, with paired t-testing on each metric.

4.11 Performance Metrics

For each trial, the recorded performance metrics included Route Delivery Ratio, Average Latency, Average Hop Count, Average Energy Cost, Average Disaster Risk, and the Average Number of High-Risk Nodes in the selected path. Results were summarized using the mean ± standard deviation. Paired t-tests were then used at each disaster severity level to compare GNN-RL with LSTM-RL and, in the ablation study, with the corresponding no-GNN condition.

4.12 Computational Cost Evaluation

The number of trainable parameters, training time, and inference latency were measured for both the LSTM and GNN models with all measurements collected on identical hardware in the same experimental session to ensure a fair comparison between the two models.

5. Experimental Results and Analysis

5.1 Overview

All six routing models were evaluated under identical network conditions using the metrics defined in Section 4.11. Figure 2 illustrates the simulated network environment used throughout these experiments. The following subsections report baseline performance, an ablation study isolating the GNN's contribution, a confirmatory evaluation under Extreme disaster severity, and a computational cost comparison.

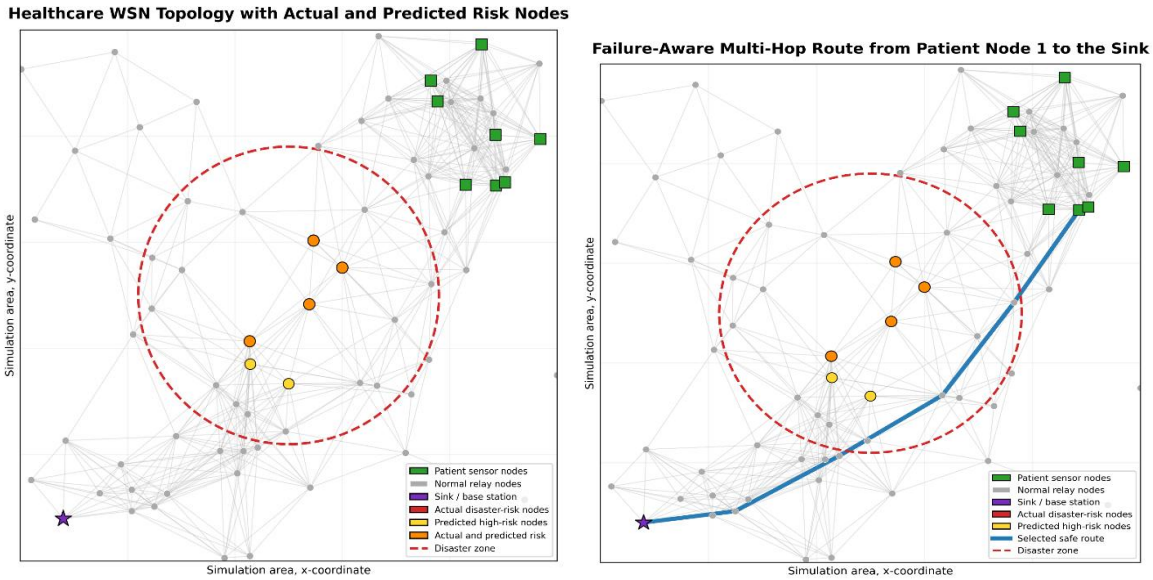


Figure 2. Simulated healthcare WSN environment. (a) Actual versus GNN-predicted high-risk nodes within the topology. (b) Example self-healing route selected by the RL agent from a patient node to the sink, avoiding the disaster zone and predicted high-risk nodes.

5.2 Baseline Routing Performance (Full Comparison, Low Severity)

Table 1. Full Model Comparison Under Low Severity (mean ± std, 30 trials)

Model	Route Delivery Ratio	Latency (ms)	Hop Count	Energy Cost	Disaster Risk
Shortest-Path	1.00±0.00	328.38±35.01	5.85±0.33	10.49±0.29	0.31±0.04
Energy-Aware	1.00±0.00	296.85±44.89	6.98±0.82	11.76±1.13	0.20±0.05
Weighted-Risk-Aware	1.00±0.00	229.00±23.66	6.43±0.39	10.85±0.44	0.15±0.04
LSTM	1.00±0.00	246.28±23.71	6.56±0.38	10.50±0.32	0.17±0.03
LSTM-RL	1.00±0.00	290.81±33.03	7.28±0.38	11.76±0.63	0.20±0.03
GNN-RL (proposed)	1.00±0.00	293.32±31.65	7.22±0.56	11.70±0.64	0.21±0.04

Shortest-path routing achieved the lowest hop count but the highest disaster risk, confirming that path length alone is a poor reliability proxy. Weighted-Risk-Aware achieved the strongest performance among conventional baselines while GNN-RL and LSTM-RL performed comparably to each other at this severity level.

5.3 Ablation Study I: Classification-Trained GNN

To isolate the GNN's specific contribution, RL routing using GNN-predicted risk was compared against RL routing using the weighted-formula risk directly (no GNN), holding the RL agent and network identical (Section 4.9).

Table 2. Ablation Results — Classification-Trained GNN (Low Severity, n=30)

Metric	No-GNN (Formula)	GNN-RL (Classification)	p-value	Significant?
Latency (ms)	255.35±34.39	299.52±29.37	<0.001	Yes — No-GNN wins
Hop Count	6.89±0.80	7.15±0.47	0.041	Yes — No-GNN wins
Energy Cost	11.27±0.80	11.66±0.54	0.009	Yes — No-GNN wins
Disaster Risk	0.172±0.032	0.227±0.038	<0.001	Yes — No-GNN wins

This unexpected result is reported in full: the classification-trained GNN did not outperform the simple weighted formula on any metric, performing significantly worse on all four metrics in Table 2. We attribute this outcome to a mismatch between the training objective and how the output is actually used. The GNN was trained as a binary classifier using thresholded labels, whereas the RL reward function uses its output as a continuous value scaled with a penalty coefficient of 150. Consequently, the model could achieve high classification accuracy while still producing probability estimates that differed substantially from the formula's true continuous risk values, introduceing noise into the reward signal without adding new information, because the pseudo-labels used to train the GNN were themselves derived from the same weighted-risk formula.

5.4 Ablation Study II: Regression-Trained GNN (Corrected)

Following this diagnosis, the GNN was retrained via MSE regression directly against the formula's continuous output (Section 3.4), and the ablation was repeated across all three severity levels.

Table 3. Ablation Results — Regression-Trained GNN (n=30 per severity)

Severity	Metric	No-GNN (Formula)	GNN-RL (Regression)	p-value	Significant?
Low	Latency (ms)	255.35±34.39	261.52±31.11	0.145	No
Low	Hop Count	6.89±0.80	6.72±0.32	0.213	No

Severity	Metric	No-GNN (Formula)	GNN-RL (Regression)	p-value	Significant?
Low	Energy Cost	11.27±0.80	11.17±0.47	0.414	No
Low	Disaster Risk	0.172±0.032	0.190±0.044	0.022	Yes (No-GNN wins)
Low	High-Risk Nodes	0.00±0.00	0.10±0.24	0.030	Yes (No-GNN wins)
Medium	Latency (ms)	266.03±43.07	275.81±33.86	0.080	No
Medium	Hop Count	6.98±0.90	6.68±0.23	0.069	No
Medium	Energy Cost	11.39±0.96	11.19±0.46	0.255	No
Medium	Disaster Risk	0.180±0.033	0.211±0.048	0.002	Yes (No-GNN wins)
Medium	High-Risk Nodes	0.004±0.023	0.083±0.253	0.097	No
High	Route Delivery Ratio	1.00±0.00	0.967±0.183	0.326	No

The regression-based correction eliminated the statistically significant differences in latency, hop count, and energy cost at both Low and Medium severity, making the GNN-RL statistically indistinguishable from those of the formula-only condition. However, the formula-only approach continued to achieve significantly lower disaster risk at both Low and Medium severity and significantly better in avoiding high-risk nodes at Low severity, suggesting the GNN did not fully reproduce the formula's implicit emphasis on disaster risk, which is assigned a direct weight of 0.25, when trained using only 73 nodes from each network instance. At High severity, GNN-RL's route delivery ratio was slightly lower than the formula-only condition (0.967 vs. 1.00), though not statistically significant at this sample size (p=0.326).

5.5 Extreme-Severity Evaluation (Regression GNN vs. LSTM-RL, n=100)

Given the ablation's diagnostic value, the corrected regression-based GNN-RL model was evaluated against LSTM-RL specifically under Extreme disaster severity across 100 trials.

Table 4. Full Comparison Under Extreme Severity (mean ± std; n=100 for LSTM-RL/GNN-RL, n=30 for others)

Model	Route Delivery Ratio	Latency (ms)	Hop Count	Energy Cost	Disaster Risk
Shortest-Path	1.00±0.00	421.26±45.87	5.91±0.37	10.61±0.35	0.38±0.04
Energy-Aware	1.00±0.00	344.73±38.55	6.98±0.64	11.57±0.84	0.25±0.06
Weighted-Risk-Aware	1.00±0.00	294.78±36.51	6.80±0.57	11.24±0.72	0.20±0.05
LSTM	1.00±0.00	319.22±37.25	7.31±0.83	11.35±1.00	0.19±0.05
LSTM-RL	0.896±0.290	357.77±69.49	8.09±1.52	12.80±2.05	0.21±0.04
GNN-RL (regression, proposed)	0.960±0.146	339.20±75.51	7.28±1.15	12.01±1.65	0.23±0.06

Table 5. Significance Test — Extreme Severity, Route Delivery Ratio

Metric	LSTM-RL	GNN-RL	p-value	Significant?
Route Delivery Ratio	0.8962	0.9600	0.0267	Yes

Across 800 routing attempts, (100 trials × 8 patient nodes), LSTM-RL failed to find a valid route in 83 attempts (10.4%), whereas GNN-RL failed in 32 attempts (4.0%). Analysis of the individual trials indicates that this advantage was concentrated under particular network conditions. In trials 3, 7, 34, 42, 50, 57, 79, and 99, LSTM-RL failed to route all eight patient nodes, while GNN-RL successfully routed most or all of them. The advantage was not universal, however; in trial 88, GNN-RL failed to route seven of eight patients

while LSTM-RL succeeded for all eight, and trial 54 resulted in complete failure for both models, suggesting some randomly generated topologies were too disconnected for any tested strategy to succeed.

5.6 Computational Cost Comparison

Table 6. Computational Cost — LSTM vs. GNN

Model	Trainable Parameters	Training Time	Single-Node Inference	Batched (74-node) Inference
LSTM	20,545	7.27 s (25 epochs)	0.391 ms	0.685 ms
GNN	1,313	0.30 s (250 epochs)	0.002 ms	0.127 ms

The GNN used approximately 15.6 times fewer trainable parameters than the LSTM and trained substantially faster despite running for 10 times more epochs while also achieving considerably lower inference latency. All measurements were collected on identical CPU hardware in the same session. Although the absolute values are hardware- specific, the relative efficiency difference between the two models is expected to remain consistent across platforms.

6. Discussion

6.1 The GNN's Contribution Is Real but Conditional, Not Universal

GNN-RL was not the strongest approach under every condition. At Low, Medium, and High severity, its performance was broadly comparable to LSTM-RL, and no statistically significant advantage was observed. However, a specific and significant benefit emerged under Extreme disaster severity, where GNN-RL achieved a higher route delivery ratio than LSTM-RL (96.0% vs. 89.6%, p=0.027), largely driven by greater resilience in a subset of severely disrupted network conditions in which LSTM-RL failed to identify any valid route. This suggests topology-aware prediction becomes most valuable under the most severe conditions, when knowledge of the broader network structure is needed to find a viable path and a node's own local history is no longer enough to provide sufficient information.

6.2 Why the Initial GNN Underperformed and What the Correction Reveals

The result reported in Section 5.3, in which the classification-trained GNN was outperformed by the formula-based condition on all four metrics, is considered as important as the positive result obtained under Extreme severity. It demonstrates that greater architectural sophistication does not automatically lead to improved performance. The alignment between a model's training objective and how its output is used by a downstream component can be as important as the model's representational capacity.

The RL reward function uses a continuous failure-risk value but, the initial GNN was trained to make a binary classification using thresholded labels. Producing a well-calibrated binary prediction was therefore not equivalent to producing the continuous risk estimate required by the RL agent, even though both relied on the same weighted-risk formula as ground truth. This shows that a model's prediction objective must match how its output is actually used downstream, not just what it is trained to approximate.

6.3 The Remaining Disaster-Risk Gap

Even after the regression-based correction, the formula-only condition continued to achieve significantly better disaster-risk at both Low and Medium severity and better high-risk-node avoidance at Low severity (Table 3). This remaining gap is attributed to the way disaster risk is represented in the two approaches. In the weighted formula, disaster risk is directly available and assigned a fixed weight of 0.25, while the GNN must learn an equivalent relationship from only 73 training nodes per network instance using the same six raw node features. Training on data from multiple network topologies rather than one at a time may help the GNN recover this relationship more fully, a direction discussed further in Section 7.2.

At High severity, GNN-RL's route delivery ratio was also slightly lower than the formula-only condition (0.967 vs. 1.00), though not statistically significant at this sample size. Consistent with the broader ablation, no severity level showed the GNN clearly outperforming the formula in isolation. Therefore, the

demonstrated benefit of the proposed framework, reported in Section 5.5, should be interpreted specifically as an advantage over LSTM-RL under Extreme disaster conditions rather than as evidence that the GNN component universally outperforms the formula-only approach.

6.4 Computational Efficiency as a Secondary, Independent Contribution

In addition to routing performance, the GNN's substantially lower trainable parameter count and reduced inference latency represent a separate practical advantage over the LSTM baseline, independent of the routing-performance comparison. A smaller and faster model is more suitable for deployment, updating, and retraining in resource-constrained sensor-network environments, where computational capacity, memory, and energy availability may be limited.

7. Limitations and Future Work

7.1 Limitations

This study relies on simulation rather than real-world deployment. Although simulation enables controlled, repeatable experimentation, it cannot fully reproduce hardware limitations, unpredictable wireless interference, patient mobility, or other operational conditions present in real healthcare sensor networks.

Both GNN variants were trained using pseudo-labels or target values derived from a hand-crafted weighted-risk formula rather than real, labeled node-failure data. This is the study's most significant limitation as it restricts what the GNN can be demonstrated to learn. As shown in Sections 5.3 and 5.4, a model trained to approximate a predefined formula cannot straightforwardly exceed the quality of the signal provided by that formula.

Additional limitations include: the Extreme severity parameters were derived by proportionally scaling the High-severity settings rather than calibrating against real disaster telemetry. In addition, the RL component used a relatively simple tabular Q-learning approach. More advanced reinforcement-learning algorithms, including Deep Q-Networks, Proximal Policy Optimization, and Actor-Critic methods [27]–[29], may produce further improvements. Security, privacy, trust management, and scalability beyond the 74-node network were also outside the scope of this study.

7.2 Future Work

The most important direction for future work is validation using real-world, failure-labeled Healthcare WSN or Wireless Body Area Network datasets, which would remove the pseudo-label limitation identified in this study and could allow the GNN to learn relationships not already encoded in a hand-crafted weighted formula. Training the GNN across multiple network topologies simultaneously, rather than on one topology at a time, may also help address the remaining disaster-risk gap identified in Section 6.3.

Further work may investigate more expressive GNN architectures, including Graph Attention Networks [19]. Although this study tested a Graph Attention Network and found that it did not resolve the underlying training-label mismatch, the architecture may become more valuable when trained using authentic failure-labeled data. Other future directions include dynamic disaster models with time-varying spread patterns, security-aware routing, and evaluation on larger-scale network configurations.

8. Conclusion

This paper developed a regression-based GNN-RL routing framework for disaster-aware HWSNs. The initial classification GNN did not improve routing over the direct weighted-risk formula. Examination of the ablation results showed that the binary training objective did not match the continuous risk penalty used by the Q-learning reward. Retraining the model through regression removed the significant differences in latency, hop count, and energy cost, although formula-based routing continued to produce lower disaster-risk exposure in some conditions.

The clearest benefit of GNN-RL appeared under Extreme disaster severity. Across 100 trials, GNN-RL achieved a route delivery ratio of 96.0%, compared with 89.6% for LSTM-RL, and the difference was statistically significant ($p = 0.027$). The GNN also used fewer trainable parameters and achieved lower inference latency than the LSTM model. These results indicate that topology-aware risk estimation can

improve resilience under severe network disruption, but the benefit was not observed consistently at every severity level. Validation using real failure-labelled healthcare sensor data remains necessary.

Data Availability Statement

The experiments in this study use simulated network data. [CONFIRM WHETHER CODE/DATA WILL BE SHARED AND ADD THE FINAL AVAILABILITY STATEMENT.]

Funding

[CONFIRM FUNDING STATUS BEFORE JOURNAL SUBMISSION.]

Conflicts of Interest

[CONFIRM CONFLICT-OF-INTEREST STATEMENT BEFORE JOURNAL SUBMISSION.]

Ethical Approval and Consent to Participate

Not applicable.

References

[1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, "Wireless sensor networks: A survey," Computer Networks, vol. 38, no. 4, pp. 393–422, 2002.

[2] J. Yick, B. Mukherjee, and D. Ghosal, "Wireless sensor network survey," Computer Networks, vol. 52, no. 12, pp. 2292–2330, 2008.

[3] M. S. Akbar et al., "Wireless Body Area Sensor Networks: Survey of MAC and Routing Protocols for Patient Monitoring under IEEE 802.15.4 and IEEE 802.15.6," *Sensors*, vol. 22, no. 21, Art. no. 8279, 2022.

[4] L. Zhong et al., "Technological Requirements and Challenges in Wireless Body Area Networks for Health Monitoring," Sensors, 2022.

[5] M. Yaghoubi et al., "Wireless Body Area Network: A Survey on Architecture, Technologies, Energy Consumption, and Security," Journal of Sensor and Actuator Networks, 2022.

[6] G. H. Adday et al., "Fault Tolerance Structures in Wireless Sensor Networks," Sensors, 2022.

[7] H. Alwan and A. Agarwal, "A Survey on Fault Tolerant Routing Techniques in Wireless Sensor Networks," International Conference on Sensor Technologies and Applications, 2009.

[8] R. Dogra et al., "ESEERP: Enhanced Smart Energy Efficient Routing Protocol for Wireless Sensor Networks," Sensors, 2022.

[9] L. Yang et al., "Energy Efficient Cluster-Based Routing Protocol for Wireless Sensor Networks," Scientific Reports, 2024.

[10] D. Godfrey et al., "An Energy-Efficient Routing Protocol with Reinforcement Learning in Wireless Sensor Networks," Sensors, 2023.

[11] S. Thakur et al., "AI-Driven Energy-Efficient Routing in IoT-Based Wireless Sensor Networks," Sensors, 2025.

[12] M. A. Alsheikh et al., "Machine Learning in Wireless Sensor Networks: Algorithms, Strategies, and Applications," IEEE Communications Surveys & Tutorials, 2014.

[13] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.

[14] F. A. Gers, J. Schmidhuber, and F. Cummins, "Learning to Forget: Continual Prediction with LSTM," Neural Computation, 2000.

[15] A. Graves, Supervised Sequence Labelling with Recurrent Neural Networks. Springer, 2012.

[16] Z. Wu et al., "A Comprehensive Survey on Graph Neural Networks," IEEE Transactions on Neural Networks and Learning Systems, 2021.

[17] F. Scarselli et al., "The Graph Neural Network Model," IEEE Transactions on Neural Networks, 2009.

[18] T. N. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," International Conference on Learning Representations, 2017.

[19] P. Veličković et al., "Graph Attention Networks," International Conference on Learning Representations, 2018.

[20] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive Representation Learning on Large Graphs," NeurIPS, 2017.

[21] Y. Shen et al., "Graph Neural Networks for Scalable Radio Resource Management: Architecture Design and Theoretical Analysis," IEEE Journal on Selected Areas in Communications, 2021.

[22] M. Eisen and A. Ribeiro, "Optimal Wireless Resource Allocation with Random Edge Graph Neural Networks," IEEE Transactions on Signal Processing, 2020.

[23] Z. Wang, M. Eisen, and A. Ribeiro, "Learning Decentralized Wireless Resource Allocations with Graph Neural Networks," IEEE Transactions on Signal Processing, 2022.

[24] H. Yang et al., "Knowledge-Driven Resource Allocation for D2D Networks: A WMMSE Unrolled Graph Neural Network Approach," 2023.

[25] Y. Dai et al., "A Survey of Graph-Based Resource Management in Wireless Networks," 2025.

[26] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. MIT Press, 2018.

[27] C. J. C. H. Watkins and P. Dayan, "Q-Learning," Machine Learning, vol. 8, pp. 279–292, 1992.

[28] V. Mnih et al., "Human-Level Control Through Deep Reinforcement Learning," Nature, vol. 518, pp. 529–533, 2015.

[29] J. Schulman et al., "Proximal Policy Optimization Algorithms," arXiv preprint arXiv:1707.06347, 2017.

[30] V. R. Konda and J. N. Tsitsiklis, "Actor-Critic Algorithms," NeurIPS, 2000.

[31] J. A. Boyan and M. L. Littman, "Packet Routing in Dynamically Changing Networks: A Reinforcement Learning Approach," NeurIPS, 1994.

[32] Y. Wang et al., "Reinforcement Learning for Energy-Saving and Routing Optimization in Wireless Sensor Networks," IEEE Access, 2024.

[33] R. A. Khan et al., "The State-of-the-Art Wireless Body Area Sensor Networks: A Survey," International Journal of Distributed Sensor Networks, 2018.

[34] W. Jian et al., "Systematic Survey on Data Security in Wireless Body Area Networks," Frontiers in Medicine, 2024.

[35] A. Ojha and B. Gupta, "Evolving Landscape of Wireless Sensor Networks: A Survey of Trends, Timelines, and Future Perspectives," Discover Applied Sciences, 2025.